

NEXT GEN HEALTHCARE MONITORING SYSTEM WITH AI POWERED IOT

A PROJECT REPORT

Submitted by

HARIPRIYA S V

KAVYA V P

SHIVANI S M

VIBESHA V

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VELLICHANTHAI

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ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report **“NEXT GEN HEALTHCARE MONITORING SYSTEM WITH AI POWERED IOT”** is the Bonafide work of **“HARIPRIYA S V (960221243023), KAVYA V P (960221243029), SHIVANI S M (960221243043), VIBESHA V (960221243050)”** who carried out the project work under my supervision.

SIGNATURE

Dr. T. Sunitha M.E., Ph.D.

HEAD OF THE DEPARTMENT

Professor, Department of AI &DS
Arunachala College of Engineering
for Women
Vellichanthai – 629 203

SIGNATURE

Dr. T. Sunitha M.E., Ph.D.

SUPERVISOR

Professor, Department of AI&DS
Arunachala College of Engineering
for Women
Vellichanthai- 629 203

Submitted to B.Tech. Project Viva – Voce Examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Effective management of chronic diseases necessitates continuous health monitoring and timely interventions to prevent complications and improve patient outcomes. This project introduces an IoT-based remote health monitoring system that leverages the Arduino Nano microcontroller, MAX30100 pulse oximeter sensor, LM35 temperature sensor, and the HC-05 Bluetooth module to collect and transmit vital patient health data in real time. The system is designed to monitor key health parameters such as blood oxygen saturation (SpO_2), heart rate, and body temperature. Patient-side data acquisition is facilitated through lightweight hardware, while communication with a mobile application is established via Bluetooth, ensuring reliable and low-power connectivity. The mobile app acts as a bridge between the patient and healthcare provider, allowing real-time data visualization and automatic uploads to a cloud-based storage service for long-term tracking and remote accessibility. An AI-based analytics engine integrated into the cloud provides intelligent health insights by identifying abnormal patterns in physiological data. This solution addresses the gaps in traditional chronic disease management systems by offering a low-cost, portable, and accessible health monitoring platform. It is particularly beneficial for elderly individuals, patients with mobility limitations, and those in rural or underserved regions, enabling improved continuity of care and better health outcomes.

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LIST OF ABBREVIATIONS

IoT	Internet of Things
AI	Artificial Intelligence
API	Application Programming Interface
IFTTT	If This Then That
HIPAA	Health Insurance Portability and Accountability Act
GDPR	General Data Protection and Regulation
GPIO	General Purpose Input/Output
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
ECG	Electrocardiogram
PHP	Hypertext Preprocessor
HTML	HyperText Markup Language
SMS	Short Message Service
GSM	Global System for Mobile Communication
AWS	Amazon Web Services
PPG	Photoplethysmogram
CNNs	Convolutional Neural Networks
RNNs	Recurrent Neural Networks
COPD	Chronic Obstructive Pulmonary Disease
PWM	Pulse Width Modulation
CPU	Central Processing Unit
ADC	Analog-to-Digital Converter
IDE	Integrated Development Environment
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver-Transmitter

TX	Transmitter
RX	Receiver
LED	Light Emitting Diode
IR	Infrared
SDA	Serial Data Line

CHAPTER I

INTRODUCTION

The rising prevalence of chronic diseases such as cardiovascular disorders, and respiratory conditions demands continuous monitoring of vital health parameters. Traditional healthcare systems, which rely on periodic clinical visits, often delay critical health interventions especially for individuals in remote or underserved areas. With the advancement of technologies like the Internet of Things (IoT) and Artificial Intelligence (AI), it is now possible to shift from reactive to proactive healthcare. This project proposes a low-cost, portable, and intelligent health monitoring system that uses Arduino Nano, biomedical sensors, and Bluetooth technology to track real-time health data. The system connects to an Android application and cloud storage, while AI algorithms analyse data to detect anomalies and offer timely health recommendations. This solution aims to improve accessibility, early diagnosis, and management of chronic diseases, particularly in resource-limited settings.

1.1 Problem Statement

Chronic diseases such as cardiovascular disorders, and respiratory ailments require regular monitoring of vital health parameters. However, traditional healthcare systems are mostly reactive and rely on periodic visits to clinics or hospitals. This leads to delays in the detection of critical changes in a patient's health. Moreover, individuals residing in remote or rural areas often face barriers in accessing timely medical services due to geographic, infrastructural, and financial constraints. Existing wearable health trackers and remote monitoring systems may not be integrated, affordable, or intelligent enough to provide actionable insights or allow direct communication with healthcare providers. Therefore, a low-cost, intelligent, and easily deployable health monitoring

solution is urgently needed to support chronic disease management and remote healthcare delivery.

1.2 Background

The integration of emerging technologies like the Internet of Things (IoT) and Artificial Intelligence (AI) is transforming the landscape of modern healthcare. IoT devices such as sensors and microcontrollers can collect real-time physiological data and communicate it wirelessly, enhancing the ability to remotely monitor patients outside traditional healthcare settings. With microcontrollers like the Arduino Nano and communication modules such as HC-05 Bluetooth, it is now feasible to create compact and efficient health monitoring devices. Additionally, AI algorithms can analyze collected data to detect abnormal trends, issue alerts, and even suggest medical actions based on predictive models. This technological synergy enables healthcare to shift from episodic care to continuous, data-driven, and proactive patient management.

1.3 Need for the Project

This project addresses the need for a reliable and intelligent remote health monitoring system by combining low-cost hardware and AI-enabled software. The motivation for this project stems from:

- The global increase in chronic disease prevalence.
- The inadequacy of in-hospital monitoring for long-term conditions.
- The lack of personalized insights in many existing monitoring devices.
- The growing demand for telemedicine and home-based healthcare solutions.

By offering a Bluetooth-enabled system that transmits real-time health data to an Android application and integrates AI-based insights, the project aims to bridge the healthcare accessibility gap.

1.4 Objectives of the Project

- To design a portable health monitoring system using Arduino Nano, MAX30100 sensor, LM35 temperature sensor, and HC-05 Bluetooth module.
- To develop an Android-based mobile application for patients to view live health data.
- To implement a cloud-based storage system for recording and accessing patient history.
- To integrate AI-based algorithms for detecting anomalies in vital signs and providing health recommendations.
- To enable real-time alerts for critical conditions to improve early diagnosis and reduce medical response time.

1.5 Scope of the Project

The system focuses on the continuous measurement of key vital signs—oxygen saturation (SpO₂), heart rate, and body temperature. The use of Bluetooth allows real-time communication between the monitoring device and a mobile application. Data is securely uploaded to a cloud database where AI-based analytics provide personalized health insights and early warnings. The system is especially useful for chronic patients, the elderly, and those in medically underserved areas. Future expansions could include additional sensors, machine learning models, and wider platform compatibility.

CHAPTER II

LITERATURE REVIEW

2.1 IoT-Based Remote Patient Monitoring – Shaik et al., 2023

Description:

In this study, Shaik et al. propose an IoT-based architecture for remote healthcare monitoring. The system uses microcontrollers, sensors, and cloud integration to track vitals like heart rate, SpO₂, and body temperature. Data is wirelessly transmitted to a cloud database accessible by healthcare professionals, enabling real-time monitoring and reducing hospital visits.

A key feature is the use of AI for anomaly detection. Machine learning models trained on patient data identify irregularities and alert clinicians for early intervention, aiding chronic disease management.

The paper also highlights the use of encryption to secure sensitive health data during transmission and storage.

Challenges and Limitations:

- Reliance on Wi-Fi limits usability in low-connectivity areas.
- High-cost components and paid cloud services reduce suitability for low-resource settings.
- Power consumption issues are not fully addressed.
- AI lacks personalization and adaptive learning.

Potential Improvements:

- Use cost-effective hardware like Arduino Nano and HC-05 for short-range communication.
- Adopt Firebase for lightweight, real-time data storage.
- Implement edge AI on mobile apps to minimize cloud dependency.

- Enhance AI with trend analysis and adaptive learning.
- Optimize power usage with low-power sensors and sleep modes.

2.2 Development of IoT-Based Sleep Apnea Monitoring – Dhruba et al., 2022

Description:

Dhruba et al. proposed an IoT-based system for diagnosing and monitoring sleep apnea. It used an Arduino Uno with a MAX30100 pulse oximeter to track blood oxygen and heart rate, and a piezoelectric sensor to monitor chest movements. Data was sent via HC-05 Bluetooth to an Android app for real-time visualization.

Designed for home use, the system offered a low-cost alternative to hospital sleep tests. The app included a simple interface that alerted users when oxygen levels dropped, indicating possible apnea events. It emphasized low power use and offline functionality for remote environments.

Challenges and Limitations:

- Limited to sleep apnea-related metrics.
- No cloud storage for long-term analysis.
- Lacked AI; alerts were based on fixed rules.
- Overnight power management was not optimized.
- No clinician interface for medical follow-up.

Potential Improvements:

- Extend monitoring to other vitals for broader chronic care.
- Use Firebase for cloud storage and access by doctors.
- Add AI for personalized alerts and trend prediction.
- Switch to Arduino Nano for better portability.
- Include scheduled alerts and daily health summaries for users.

2.3 Design and Implementation of a Smart Health Monitoring System – Vaibhav & Ahmad, 2018 (Springer)

Description:

Vaibhav and Ahmad's work presents a practical IoT-based health monitoring system using Arduino Uno, LM35 temperature sensor, and a heart rate sensor. Data is transmitted via the HC-05 Bluetooth module to a mobile app built with MIT App Inventor, allowing users to view real-time temperature and pulse rate readings.

The system emphasizes cost-effectiveness, portability, and ease of use, making it suitable for rural and resource-limited settings. The prototype demonstrated accurate real-time monitoring and served as an effective tool for basic health screening at home.

Challenges and Limitations:

- No cloud support—data wasn't stored for long-term access.
- No interface for doctors to review patient data.
- Only rule-based alerts, no AI or adaptive learning.
- Bluetooth communication lacked encryption.
- No support for long-term trend visualization.

Potential Improvements:

- Integrate Firebase for cloud storage and remote access.
- Add AI for personalized alerts and trend prediction.
- Include more vitals like SpO₂ for broader monitoring.
- Develop AI-driven suggestions using historical data.
- Switch to Arduino Nano for a more compact, wearable design.
- Implement Bluetooth encryption to enhance data security.

2.4 Cloud-Based Health Monitoring System Using IoT – K. S. Ravikumar et al., 2020 (IEEE)

Description:

In this IEEE-backed research, Ravikumar et al. introduced a cloud-enabled IoT system for remote patient monitoring. Using ESP8266 (NodeMCU) with MAX30100 and LM35 sensors, the system tracked SpO₂, heart rate, and temperature, transmitting data via Wi-Fi to ThingSpeak for real-time and historical access. Alerts were sent using IFTTT via SMS or email.

Designed for chronic care, the system allowed healthcare providers to monitor multiple patients remotely, improving response times and long-term oversight.

Challenges and Limitations:

- No mobile app limited real-time access for patients.
- Alerts were static and lacked personalization.
- ThingSpeak has data rate and scalability constraints.
- No two-way communication for doctor-patient interaction.
- Minimal focus on data security.

Potential Improvements:

- Use Firebase for scalable, secure, and real-time data syncing.
- Create an Android app for live tracking, alerts, and history.
- Add AI for personalized, adaptive alerts.
- Implement encryption and authentication to protect data.
- Upgrade to ESP32 for better performance and Bluetooth support.

2.5 AI-Integrated Wearable Health Monitoring System – N. Karthikeyan et al., 2021 (Springer)

Description:

Karthikeyan et al. developed an AI-integrated wearable health monitoring system using an ESP32 microcontroller with MAX30102 (SpO₂, heart rate) and LM35 (temperature). Data is sent via Wi-Fi to the cloud for continuous monitoring.

The standout feature is an AI backend using a Support Vector Machine (SVM) to classify patient status (Normal, Warning, Critical). Alerts and health insights are delivered through a Firebase-connected mobile app. A visual dashboard helps doctors track trends and spot anomalies, especially useful for elderly and chronic disease patients.

Challenges and Limitations:

- SVM may not scale well to broader datasets without retraining.
- Battery optimization for the wearable wasn't detailed.
- Relies on Wi-Fi—limited usability in low-connectivity areas.
- Lacked role-based access in Firebase, raising privacy concerns.

Potential Improvements:

- Use LSTM for better time-series prediction in chronic care.
- Apply low-power modes to extend battery life.
- Add Bluetooth or offline data logging for rural use.
- Implement encryption and access control in Firebase.
- Integrate video consultations to build a full telemedicine solution.

2.6 IoT-Based Real-Time Health Monitoring and Alert System – A. S. Pawar et al., 2019 (IJREST)

Description:

Pawar et al. introduced a real-time IoT health monitoring system using ESP8266 with sensors like MAX30100 (SpO₂, heart rate), DHT11 (temperature/humidity), and ECG electrodes. Data is sent via Wi-Fi to a web dashboard (HTML/PHP/JavaScript), and alerts are delivered through a GSM-based SMS system.

Designed for elderly and chronic patients, the system supports continuous monitoring and quick emergency response via real-time alerts.

Challenges and Limitations:

- No mobile app limits on-the-go access.
- Lacks cloud storage for long-term data analysis.
- Alerts use static thresholds—no AI or predictive analytics.
- ESP8266 restricts sensor expansion and complex processing.
- SMS alerts may incur recurring costs and depend on network availability.

Potential Improvements:

- Add Firebase or AWS for scalable, cloud-based data storage.
- Integrate AI/ML for health trend analysis and predictive alerts.
- Develop a mobile app with live data and two-way communication.
- Enable offline storage or Bluetooth for low-connectivity areas.
- Implement power-saving modes to extend device battery life.
- Expand alerts to push notifications, email, or app-based messages.

2.7 IoT and AI-Based Smart Healthcare System for Chronic Disease Management – R. Patel et al., 2021 (Springer)

Description:

Patel et al. proposed an IoT and AI-based smart healthcare system for continuous monitoring of chronic disease patients (heart disease, diabetes, hypertension). Using an ESP32 microcontroller and sensors for heart rate, SpO₂, temperature, and blood pressure, data is sent via Wi-Fi to Microsoft Azure cloud for storage and AI-driven analysis. The system employs regression and decision tree models to predict health risks and generate early warnings.

A mobile app enables patients to track health data and receive alerts, while doctors can remotely review trends and intervene as needed.

Challenges and Limitations:

- Data reliability may be affected by low-cost sensors.
- AI models need better optimization and disease-specific tuning.
- Data privacy risks exist despite secure transmission protocols.
- Cloud scalability and cost could be issues with large user bases.
- Dependence on stable internet limits use in rural/remote areas.

Potential Improvements:

- Use higher-quality sensors and calibration for accuracy.
- Adopt advanced AI (deep learning) to improve predictions.
- Implement decentralized storage like blockchain for security.
- Enhance mobile app with offline features and interactivity.
- Apply low-power modes and local data storage for energy efficiency and connectivity resilience.
- Introduce edge computing for faster, real-time data processing and alerts.

2.8 A Cloud-Based Healthcare Monitoring System Using IoT and AI – M. S. Ghosh et al., 2020 (IEEE Xplore)

Description:

Ghosh et al. proposed a cloud-enabled healthcare monitoring platform targeting chronic disease patients. The system continuously tracks vitals—using pulse oximeters, temperature sensors, ECG, and blood pressure monitors—and securely transmits data to Microsoft Azure cloud. AI-powered analytics assess real-time and historical data to predict health risks like heart attacks or severe hypoxia. Alerts are sent via SMS, email, or push notifications to patients and doctors. A mobile app and web interface provide access to health metrics and trends, offering scalable, flexible remote monitoring ideal for patients in underserved or remote regions.

Challenges and Limitations:

- **Sensor data accuracy:** Variability due to sensor quality and environmental factors; e.g., ECG readings affected by motion or poor contact.
- **Data sync and latency:** Multiple sensors and cloud transmission can cause delays, risking late detection of critical events.
- **Security/privacy:** Cloud storage poses risks despite encryption and secure protocols, with potential unauthorized access.
- **Internet dependency:** Requires constant connectivity, problematic in remote areas with poor internet, risking monitoring interruptions.
- **Cloud costs:** Scaling up can be expensive, especially for continuous storage across large patient bases.

Potential Improvements:

- **Advanced sensor calibration & redundancy:** Use better sensors with self-calibration and cross-checking multiple sensors to improve reliability and reduce false alarms.

- **Edge computing:** Process data locally on devices to minimize latency and enable instant alerts without waiting for cloud analysis.
- **Blockchain for security:** Implement decentralized, immutable data storage to enhance access control, data integrity, and auditability.
- **Offline capabilities:** Local data storage during internet outages, syncing with cloud once connectivity resumes for uninterrupted monitoring.
- **Personalized AI insights:** Employ sophisticated AI that factors in patient history, lifestyle, and genetics for better predictive healthcare and early risk detection.
- **Multilingual app support:** Expand accessibility by supporting multiple languages in the patient mobile app for diverse populations.

2.9 An Intelligent Health Monitoring System Using IoT and AI for Chronic Disease – S. Kumar et al., 2019 (IEEE Xplore)

Description:

Kumar et al. developed a wearable IoT-based health monitoring system targeting chronic diseases such as diabetes, hypertension, respiratory diseases, and cardiovascular diseases. The system continuously measures vital signs including blood sugar, blood pressure, heart rate, oxygen saturation (SpO₂), and respiratory rate. Data is sent to a centralized cloud platform where AI algorithms analyze it for personalized health insights and abnormal condition detection. Supervised learning models (decision trees, SVM) classify data to identify critical health events. Alerts are sent via a mobile app to patients and via a web interface to doctors for remote monitoring and intervention.

Challenges and Limitations:

- **Sensor calibration issues:** Blood pressure and glucose sensors may produce inaccurate readings, risking false alarms or missed detections.

- **Data processing delay:** Cloud transmission introduces latency, which can delay critical health alerts in urgent situations.
- **Data privacy and compliance:** Cloud storage of sensitive data poses security risks; compliance with regulations (HIPAA, GDPR) requires strong encryption and access controls.
- **Power consumption:** Continuous monitoring and frequent data transmission drain wearable battery life quickly.
- **AI model complexity:** Predictive AI models may be difficult for healthcare professionals to interpret, limiting clinical acceptance.

Potential Improvements:

- **Sensor accuracy and robustness:** Use multi-modal sensors (e.g., combining ECG with PPG) and real-time calibration techniques that account for environmental factors to improve data reliability.
- **Edge computing:** Process data locally on devices or edge servers to reduce latency and provide faster alerts.
- **Advanced AI models:** Implement deep learning approaches (CNNs, RNNs) to better capture complex patterns and improve early detection of disease progression.
- **Blockchain integration:** Enhance data security and transparency with immutable, decentralized health records.
- **Energy-efficient designs:** Utilize low-power sensors, energy harvesting (e.g., solar), and power-saving transmission modes to extend device battery life.
- **User-friendly interfaces:** Develop intuitive mobile apps with customizable alerts, clear visualizations, and simple health summaries to increase patient engagement.

2.10 Mobile Application for Real-Time Patient Monitoring – Linkous et al., 2019 (SpringerLink)

Description:

Linkous et al. developed a mobile app integrated with wearable IoT sensors for real-time monitoring of chronic diseases such as diabetes, hypertension, and asthma. The app collects health data (heart rate, blood pressure, SpO₂, temperature) from sensors and transmits it to a centralized server and healthcare providers via the internet. Patients can view real-time metrics and historical trends through a user-friendly interface, while doctors remotely monitor multiple patients, receiving alerts for abnormal health parameters. The app includes push notifications, educational resources, reminders, and personalized health tips to enhance patient engagement. Data privacy is maintained through encrypted communication.

Challenges and Limitations:

- **Sensor calibration and accuracy:** Low-cost wearable sensors may provide inaccurate readings, leading to false alerts or missed diagnoses.
- **Connectivity and latency:** Poor or unstable internet can delay data transmission, reducing effectiveness of real-time monitoring and alerts.
- **Data privacy and security:** Sensitive health data stored on remote/cloud servers is vulnerable to unauthorized access without robust encryption and access controls.
- **Battery life:** Continuous sensor monitoring and frequent communication can rapidly drain wearable device batteries.
- **User interface and experience:** Complex or unintuitive app design may hinder patient use and reduce adherence to monitoring routines.

Potential Improvements:

- **Enhanced sensor accuracy:** Use clinically validated sensors and multi-sensor fusion to improve data reliability by compensating for errors and inconsistencies.
- **Offline functionality:** Implement offline data storage on the device during connectivity loss, syncing with cloud/servers when online again, ensuring uninterrupted monitoring.
- **AI for predictive analytics:** Integrate machine learning models to analyze real-time and historical data, detecting early warning signs and providing preventive insights.
- **Improved user engagement:** Add interactive health goals, fitness tracking, gamification, motivational notifications, and social sharing to boost adherence.
- **Multilingual and culturally appropriate features:** Support multiple languages and culturally tailored health advice for wider accessibility and inclusion.
- **Edge computing:** Enable local data processing on devices or nearby infrastructure to reduce latency and allow immediate alerts without waiting for cloud processing.

CHAPTER III

SYSTEM DESIGN AND ARCHITECTURE

3.1 Existing System

The existing systems for chronic disease monitoring include traditional healthcare and telemedicine platforms. Traditional methods rely on periodic in-person check-ups, leading to delayed detection, lack of real-time monitoring, and limited accessibility in rural areas. They also involve high costs and fragmented patient data. Telemedicine allows remote consultations but often lacks integration with real-time monitoring devices and advanced AI analysis. Privacy concerns and technology access issues further limit their effectiveness. Overall, both systems are insufficient for continuous and personalized chronic disease management.

3.1.1 Current Healthcare Systems for Chronic Disease Monitoring

Traditional healthcare systems primarily rely on periodic check-ups or in-person visits to monitor chronic diseases. In these systems, patients are typically required to visit healthcare facilities at scheduled intervals for blood tests, physical examinations, and other diagnostic measures. However, this approach has several limitations:

- **Limited Frequency of Monitoring:** The infrequent check-ups mean that any significant health changes or complications may go undetected for extended periods.
- **Lack of Real-Time Monitoring:** Without continuous monitoring, patients with chronic conditions like hypertension, diabetes, or COPD are at higher risk of sudden health deteriorations.
- **Delayed Interventions:** Due to the periodic nature of check-ups, early detection and timely interventions are often missed, which may lead to severe complications.

Challenges:

- Limited accessibility, especially for patients in rural or remote areas, where healthcare facilities are sparse.
- High healthcare costs associated with frequent hospital visits and diagnostic procedures.
- Inadequate patient engagement, as patients may not receive continuous health feedback.
- Data fragmentation where patients' health data from various visits are often not integrated, making it difficult to track long-term health trends.

3.1.2 Telemedicine Systems

Telemedicine platforms aim to address some of the above issues by allowing patients to consult healthcare providers remotely via video calls, phone calls, or messaging. These platforms often include the option to share health data from wearable devices, but the adoption of these technologies remains limited by:

- **Lack of Integration with Real-Time Health Monitoring Devices:** Many telemedicine systems do not feature continuous health data monitoring; they rely on manually submitted data or intermittent check-ins.
- **Privacy Concerns:** Patient data shared through telemedicine platforms may not always be adequately secured, especially without end-to-end encryption or proper compliance with health data regulations.
- **Limited AI-Driven Decision Making:** Existing telemedicine solutions often lack advanced artificial intelligence capabilities to analyze patient data and offer personalized recommendations for chronic disease management.

Challenges:

- **Limited AI-driven insights:** Existing systems lack AI-based predictive analytics or trend analysis, which could help in early detection of anomalies in chronic conditions.
- **Technology Barriers:** Not all patients, especially the elderly or those in rural areas, have access to smartphones, internet, or telemedicine platforms.
- **Dependency on patient self-reporting,** which can result in inaccurate data submission or poor engagement.

3.2 Proposed System

The proposed system integrates IoT and AI technologies to address the limitations of traditional and telemedicine healthcare systems. It uses sensors connected to an Arduino to continuously monitor vital signs like heart rate, SpO₂, and temperature, with real-time data uploaded to a secure cloud platform. AI algorithms analyse this data to detect anomalies and provide personalized health insights. A mobile app enables patients and doctors to track health trends, and communicate effectively. This system allows for early detection, remote access, and improved chronic disease management.

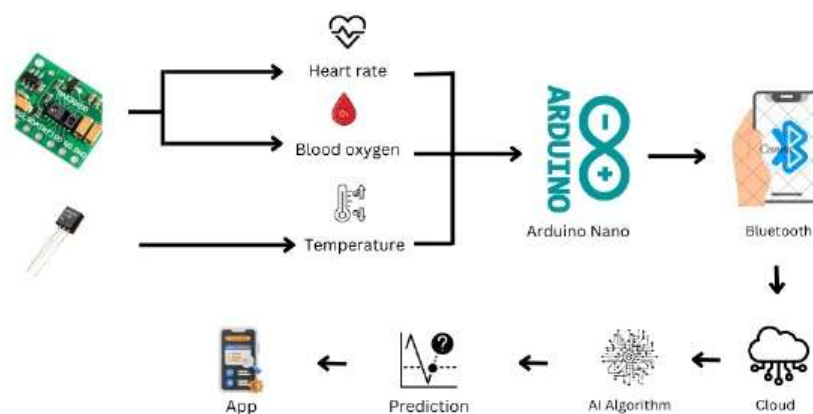


Fig 3.1 System Architecture

3.2.1 Hardware Components

MAX30100 – Pulse Oximeter and Heart Rate Sensor:

The MAX30100 is an advanced pulse oximeter and heart rate monitor sensor developed by Maxim Integrated. It is widely used in health monitoring systems to measure blood oxygen saturation (SpO_2) and heart rate (HR) in real-time. The MAX30100 sensor operates using infrared light to detect changes in blood volume during each heartbeat, which allows it to compute the oxygen levels in the blood and the heart rate.

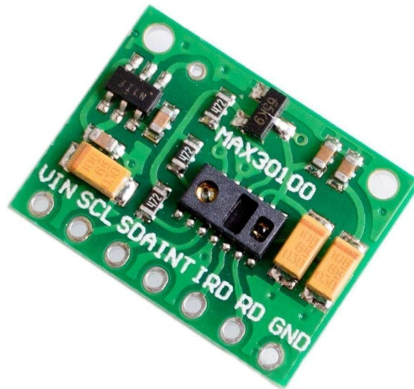


Fig 3.2 Max30100 Pulse Oximeter

- **Working Principle:**

The MAX30100 sensor consists of two main components: an LED array and a photodetector. The sensor uses a combination of infrared (IR) and red-light LEDs, which shine light onto the skin at the point where the sensor is placed (typically the fingertip or earlobe). When blood flows through the capillaries under the skin, the volume of blood changes with each pulse. This variation in blood volume causes a difference in the amount of light that is absorbed by the blood and reflected back to the photodetector.

The LED array emits light at specific wavelengths (usually red and infrared), and the photodetector measures the amount of light that is reflected back from the skin. Oxygenated and deoxygenated blood absorb different amounts of light at

these wavelengths, allowing the sensor to differentiate between them. By analysing the reflected light at these wavelengths, the MAX30100 can calculate the SpO₂ (blood oxygen saturation) level and the heart rate by observing the periodic changes in light reflection caused by the pulse.

- **Data Acquisition:**

The MAX30100 sensor continuously measures the pulse rate and SpO₂ by sampling the reflected light at regular intervals. The sensor converts the light intensity readings into digital data, which is then processed by its internal integrated circuit. It uses advanced algorithms to eliminate noise and ensure accurate readings, even in conditions where the sensor may not have perfect contact with the skin.

The resulting measurements are sent to the Arduino Nano over the I2C (Inter-Integrated Circuit) communication protocol. The I2C protocol allows for easy data transfer between the MAX30100 sensor and the Arduino Nano using only two wires (SDA for data and SCL for clock), simplifying the wiring and communication process. The Arduino Nano, acting as the central processing unit, then processes the incoming data from the MAX30100, formats it for display or further analysis, and sends it to the HC-05 Bluetooth module for wireless transmission to a mobile application.

- **Applications in Health Monitoring Systems:**

In the context of this health monitoring system, the MAX30100 sensor plays a critical role in providing continuous monitoring of cardiovascular and respiratory health. It allows the system to track both heart rate and blood oxygen levels—two key parameters for assessing the health of individuals, particularly those with chronic diseases such as heart disease, asthma, or COPD (chronic obstructive pulmonary disease).

By integrating the MAX30100 with the Arduino Nano and HC-05 Bluetooth module, the system enables wireless real-time monitoring of these vital signs. The health data is continuously transmitted to a mobile application, where healthcare providers can monitor the patient's health remotely, receive alerts in case of abnormal readings, and make data-driven decisions for patient care.

LM35 – Temperature Sensor:

- The LM35 is an analog temperature sensor that measures body temperature. It outputs a voltage that is linearly proportional to the temperature, which can be easily read by the Arduino Nano through its analog-to-digital converter (ADC).
- For every 1°C increase, the LM35 outputs a voltage increase of 10 mV, which is then converted into temperature values by the Arduino. The measured temperature data is then sent to the Bluetooth module for transmission.

Arduino Nano:

- The Arduino Nano is a small, versatile, and cost-effective microcontroller board based on the ATmega328P chip. It is a popular choice in embedded systems for prototyping and embedded applications due to its compact size, ease of use, and wide range of available libraries and community support. The Arduino Nano operates on a 5V supply voltage and is equipped with 14 digital input/output pins, 8 analog inputs, and built-in PWM outputs, making it suitable for a variety of sensor interfacing and signal processing applications.

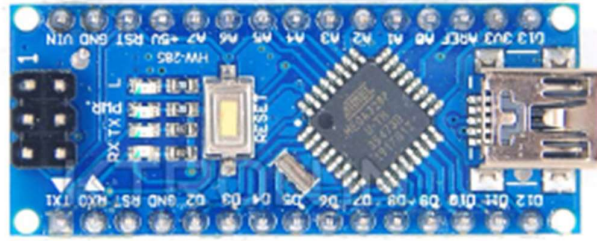


Fig 3.3 Arduino Nano

- The Arduino Nano is especially valuable in low-power applications due to its ability to run on minimal energy while still offering sufficient processing power for basic tasks. In the context of health monitoring, the Arduino Nano is an ideal choice as it can handle the continuous data acquisition from various health sensors (such as the MAX30100 Pulse Oximeter and LM35 Temperature Sensor) and transmit the results to other devices for further processing and display.
- In this specific system, the Arduino Nano serves as the central processing unit (CPU) that manages the entire health monitoring process. It interacts with the health sensors and processes the collected data by performing necessary operations, such as analog-to-digital conversion and signal filtering to ensure the data is accurate and usable.
- The MAX30100 sensor, which measures blood oxygen levels (SpO_2) and heart rate, produces data in digital form via the I2C interface. The LM35 temperature sensor, on the other hand, provides analog data that is read using the Arduino Nano's analog-to-digital converter (ADC). The Arduino reads both types of data and prepares it for transmission.
- After processing the data from the sensors, the Arduino Nano sends the processed health parameters (such as heart rate, SpO_2 levels, and body temperature) to the HC-05 Bluetooth module. The HC-05 is a Bluetooth serial communication module that facilitates wireless transmission of data. By pairing the Bluetooth module with a mobile application, the Arduino

Nano transmits the health data to the mobile device in real-time, enabling patients and healthcare professionals to monitor the health status remotely.

- The Arduino Nano is particularly beneficial for this system due to several key features:
- **Small Form Factor:** The compact design of the Arduino Nano makes it easy to integrate into portable health monitoring systems, making it an ideal solution for wearable or portable health devices.
- **Low Power Consumption:** The Arduino Nano consumes minimal power, making it well-suited for applications where battery life is a critical factor, such as wearable devices or systems that need to operate continuously over long periods.
- **Ease of Use and Flexibility:** With a large community, comprehensive documentation, and an easy-to-use development environment, the Arduino Nano simplifies the development process. The Arduino IDE allows for quick programming and uploading of code to the microcontroller, facilitating rapid prototyping and testing.
- **Interfacing with Multiple Sensors:** The Arduino Nano supports both digital and analog sensors, and with its I2C and SPI communication protocols, it is capable of connecting to a wide range of sensors and devices, making it flexible for future system expansions.
- While more powerful microcontrollers like the ESP32 offer advanced connectivity and processing capabilities, the Arduino Nano provides sufficient processing power for the task at hand, making it a cost-effective choice for health monitoring applications. Its 5V operation is compatible with many sensors, and its simplicity allows for easy integration into both the hardware and software components of the health monitoring system.

HC-05 Bluetooth Module:

- The HC-05 Bluetooth module is a widely used and highly efficient wireless communication module designed to facilitate short-range Bluetooth communication between two devices, typically between a microcontroller (like the Arduino Nano) and a mobile device. It operates on the Bluetooth 2.0 standard and supports both master and slave modes, making it flexible for a variety of communication scenarios. In this health monitoring system, the HC-05 module plays a crucial role in enabling real-time data transmission from the health monitoring system (Arduino Nano) to the mobile application.

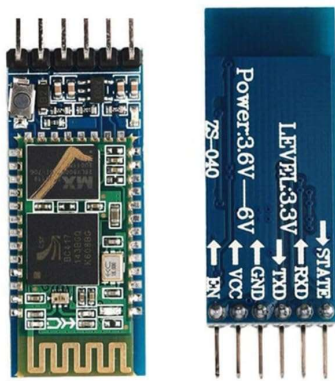


Fig 3.4 HC-05 Bluetooth Module

- The HC-05 module uses the UART (Universal Asynchronous Receiver-Transmitter) serial communication protocol to transmit data. This means that the Arduino Nano sends the health data (e.g., heart rate, SpO₂ levels, body temperature) to the HC-05 module via TX/RX pins, which are dedicated to transmitting and receiving data. The HC-05 module then wirelessly transmits this data to the paired mobile device, typically a smartphone or tablet, using Bluetooth communication.

The main advantages of the HC-05 Bluetooth module in the context of this system include:

- **Wireless Data Transmission:** The HC-05 eliminates the need for physical cables between the Arduino Nano and the mobile device, allowing for greater mobility and flexibility. This is especially important in health monitoring applications where users are on the move, such as elderly patients or people with chronic conditions who require continuous monitoring.
- **Ease of Integration:** The HC-05 is designed to be easy to integrate with microcontrollers like the Arduino Nano. It communicates via a serial connection (UART), which is simple to interface with the Arduino and does not require additional hardware. This allows for a seamless exchange of health data between the monitoring device and the mobile application.
- **Real-Time Communication:** The HC-05 ensures real-time data transmission, allowing the system to send updated health information continuously or at specified intervals to the mobile app. This real-time capability is essential in health monitoring, as it enables immediate alerts and feedback, enabling healthcare professionals to respond promptly to any abnormal health readings.
- **Secure Communication:** The HC-05 supports encryption and security protocols for Bluetooth communication, ensuring that sensitive health data is transmitted securely. This feature is vital in healthcare systems, where data privacy and patient confidentiality are paramount. Although the HC-05 does not have advanced security measures like Bluetooth 4.0 or BLE (Bluetooth Low Energy), it is suitable for basic health monitoring systems where standard security measures are adequate.
- **Cost-Effective Solution:** The HC-05 Bluetooth module is inexpensive, making it an affordable option for developing wireless communication

systems. Given that health monitoring systems often involve multiple devices and sensors, using a cost-effective communication module like the HC-05 helps keep the overall cost of the system low, making it more accessible for a larger user base, especially in resource-constrained settings.

- **Compatibility with Android/iOS Applications:** The HC-05 module is compatible with a wide range of mobile applications on both Android and iOS platforms. This ensures that healthcare professionals and patients can easily connect the health monitoring system to their devices, using any commonly available mobile application or a custom-designed app.

Use in the System:

- In the health monitoring system, the HC-05 Bluetooth module facilitates the crucial role of data transmission between the Arduino Nano and the mobile application. Once the Arduino processes the health data from the sensors (MAX30100 and LM35), the HC-05 module transmits this information wirelessly to the mobile device for visualization and analysis. This wireless data flow eliminates the need for cumbersome physical connections, providing flexibility and ease of use for the patients and healthcare providers alike.
- Additionally, when a patient's health readings (heart rate, blood oxygen levels, and body temperature) exceed predefined thresholds, the HC-05 module allows the system to send real-time alerts to the patient. These alerts can prompt immediate medical action, such as recommending an appointment with a healthcare professional or notifying emergency services in critical situations.

3.2.2 Working of the System

The proposed health monitoring system integrates hardware, software, cloud infrastructure, and artificial intelligence to offer a comprehensive solution for continuous, real-time tracking of patient health parameters. This system is particularly beneficial for individuals with chronic conditions, enabling early detection of health issues and facilitating timely medical interventions.

3.2.2.1 Sensor-Based Data Collection

The operation of the system begins with the acquisition of physiological data using a set of biomedical sensors connected to an Arduino Nano microcontroller. The MAX30100 Pulse Oximeter is used to monitor the patient's heart rate and blood oxygen saturation (SpO_2), while the LM35 temperature sensor measures body temperature. These sensors produce analog signals, which are processed and digitized by the Arduino Nano. The microcontroller also performs preliminary filtering and formatting of the data to prepare it for transmission.

3.2.2.2 Wireless Data Transmission

After data processing, the Arduino Nano sends the information to the HC-05 Bluetooth module. This module establishes a wireless communication link between the microcontroller and the patient's mobile device. The Bluetooth connection allows the system to transmit data in real time without the need for cables or complex network setups, thus ensuring portability and ease of use for patients.

3.2.2.3 Mobile Application Interface

On the patient's mobile device, a dedicated application receives the Bluetooth-transmitted data and serves as the primary interface for health monitoring. The application displays real-time values for heart rate, SpO_2 levels, and body temperature, allowing users to stay informed about their health status at all times.

In addition to live monitoring, the app maintains a historical record of health metrics and provides visualizations such as trend graphs and daily averages. The application also includes alert functionalities that notify the user when any metric exceeds safe thresholds, thereby prompting timely action.

3.2.2.4 Cloud Storage and Remote Monitoring

The system further integrates cloud storage through Firebase, which acts as a real-time backend for data synchronization and remote access. The mobile application uploads health data to Firebase, making it accessible to healthcare providers for continuous monitoring. Firebase ensures that the data is stored securely with encryption protocols and access control mechanisms that comply with healthcare privacy standards such as HIPAA. This remote monitoring capability is particularly useful in telemedicine scenarios, where clinicians can track patient conditions without requiring physical visits.

3.2.2.5 Anomaly Detection Using Random Forest

To enhance the reliability of health monitoring, the system incorporates a machine learning model based on the Random Forest algorithm. This ensemble learning method is used for real-time anomaly detection. It classifies incoming data as normal or abnormal by referencing patterns learned during the training phase. For instance, an irregular heart rate or a sudden drop in SpO₂ levels could be flagged as anomalies, indicating possible conditions such as arrhythmia or respiratory distress. The use of Random Forest ensures robust classification, even in the presence of noisy or partial data.

3.2.2.6 Predictive Analytics with Extreme Learning Machine (ELM)

In addition to detecting anomalies, the system employs the Extreme Learning Machine (ELM) algorithm to provide predictive analytics. ELM is a high-speed neural network model suitable for handling large datasets with minimal computational overhead. By analysing historical data, ELM can predict future

health trends and risks. For example, a continuous increase in heart rate or body temperature over several days may be indicative of an impending cardiovascular issue or infection. The model anticipates such developments and issues predictive alerts, allowing patients and healthcare providers to take preventive action before the condition becomes critical.

3.2.3 Advantages of the Proposed System

The proposed IoT-based, AI-powered healthcare system offers numerous advantages, both for patients and healthcare providers, particularly in the management of chronic diseases:

1. Continuous and Real-Time Monitoring:

- Unlike traditional healthcare systems that rely on intermittent check-ups, this system offers continuous monitoring of vital health parameters, ensuring that no changes go unnoticed. Heart rate, blood oxygen levels, and body temperature can be monitored 24/7, providing real-time insights into a patient's health status.

2. Early Detection of Health Anomalies:

- The integration of AI algorithms enables the early detection of health deviations, even before the patient experiences noticeable symptoms. For instance, a slight drop in SpO₂ levels or a fever could indicate the onset of an infection, allowing the system to notify both the patient and healthcare provider immediately, thereby facilitating early intervention.

3. Personalized Healthcare:

- Through continuous monitoring and the use of AI, the system can offer personalized health recommendations tailored to an individual's health data. This personalized approach improves patient engagement and ensures

that patients receive the right guidance based on their unique health conditions and trends.

4. Remote Access and Convenience:

- Patients, particularly those in rural or remote areas, benefit from remote healthcare supervision. Healthcare providers can monitor patient health data in real-time, even when they are miles away. This reduces the need for frequent hospital visits, saving time and cost for both patients and healthcare providers.

5. Enhanced Patient-Doctor Interaction:

- The mobile app provides an effective channel for communication between patients and healthcare professionals. Doctors can receive real-time health updates and alerts, enabling them to respond promptly. Virtual consultations can be conducted through the app, making healthcare more accessible and convenient for patients who may have difficulty visiting hospitals.

6. Increased Patient Adherence:

- The system's AI-powered reminders for medication and health interventions, along with real-time feedback on health parameters, encourage patients to adhere to their prescribed treatments and lifestyle changes. This improves the overall effectiveness of chronic disease management and reduces the risk of complications.

7. Data-Driven Decision Support:

- By analysing the vast amount of health data collected from patients, the system provides evidence-based insights that assist healthcare providers in making data-driven decisions. Doctors can rely on the system to monitor

trends in the patient's health, ensuring they make timely and informed decisions.

8. Scalability and Flexibility:

- The system is designed to be scalable and flexible, meaning it can easily incorporate additional sensors or integrate with other healthcare platforms as technology evolves. The AI algorithms can also be upgraded or enhanced to accommodate new medical findings, ensuring the system remains relevant and effective over time.

9. Cost Efficiency:

- By reducing the need for frequent hospital visits and enabling remote healthcare supervision, the proposed system offers cost savings for both patients and healthcare providers. Additionally, by catching health issues early, it reduces the need for expensive emergency interventions or hospitalizations.

10. Improved Patient Outcomes:

- Ultimately, the most significant advantage of the proposed system is its potential to improve patient outcomes. By enabling continuous monitoring, early detection, and personalized interventions, the system empowers patients to take proactive control of their health, leading to better disease management, fewer complications, and enhanced quality of life.

CHAPTER IV

RESULTS AND DISCUSSIONS

This chapter presents the detailed results and discussions related to the implementation of the health monitoring system, utilizing the revised hardware components: Arduino Nano, MAX30100 sensor, LM35 temperature sensor, and the HC-05 Bluetooth module. The figures presented below illustrate the user interface of the application, along with the corresponding prediction results generated for various individual users and test cases. These outputs demonstrate how the system responds to different user inputs and test cases.

User 1:

User id: kavyavishnu1812@gmail.com

- The pulse rate, oxygen saturation level, and body temperature readings were carefully collected from User 1 using the healthcare monitoring system.

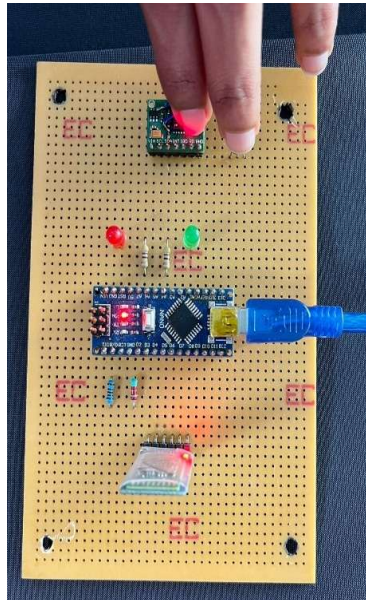


Fig 4.1: Data Collection-User 1

- After analysing the data, it was observed that all the vital parameters fall within the normal healthy range.

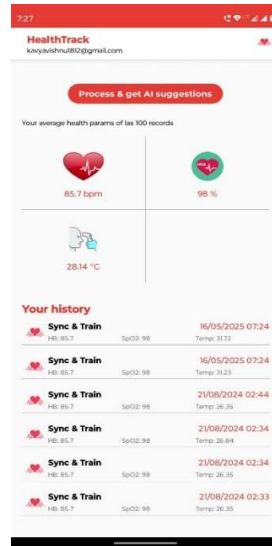


Fig 4.2: User Interface-User 1

- Indicating that the individual does not show any signs or symptoms of a specific disease or medical abnormality at this time.

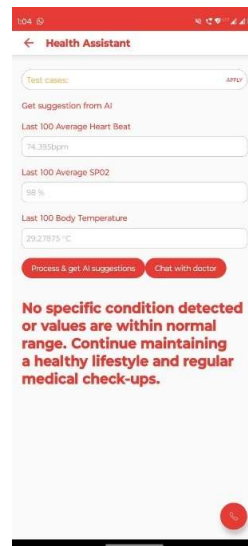


Fig 4.3: Prediction Result-User 1

User 2:

User id: vibi2472004@gmail.com

- The pulse rate, oxygen saturation level, and body temperature readings were carefully collected from User 1 using the healthcare monitoring system.

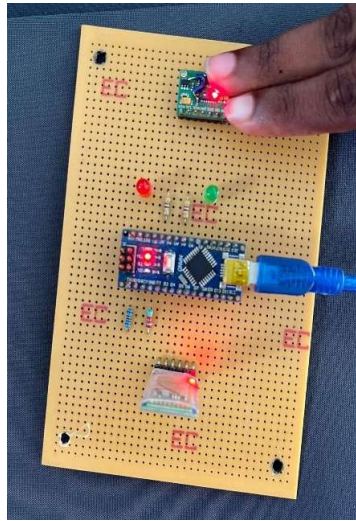


Fig 4.4: Data Collection-User 2

- After analysing the data, it was observed that all the vital parameters fall within the normal healthy range.

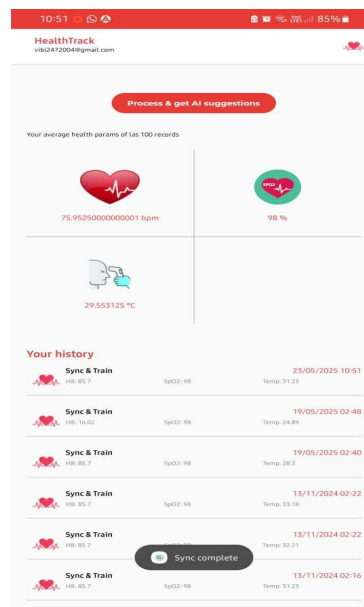


Fig 4.5 User Interface-User 2

- Indicating that the individual does not show any signs or symptoms of a specific disease or medical abnormality at this time.

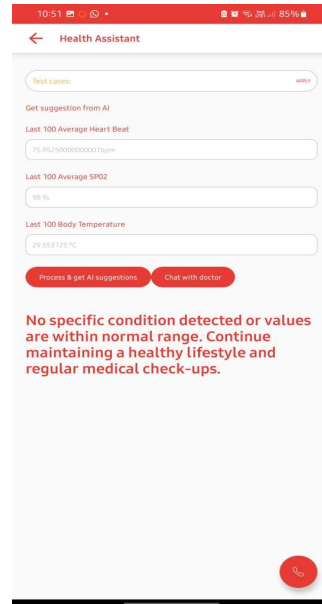


Fig 4.6: Prediction Result-User 2

Test Cases:

The following images illustrate individual test cases where three different medical conditions have been detected and classified. Each image corresponds to a specific disease based on the collected vital parameters and diagnostic analysis. The conditions represented include heart stroke, which may be identified through abnormal heart rate and oxygen patterns; arrhythmias, characterized by irregular pulse rhythms; and pneumonia, which can be indicated by abnormal oxygen saturation levels and elevated body temperature. These examples highlight the system's ability to differentiate between various health conditions using real-time patient data.

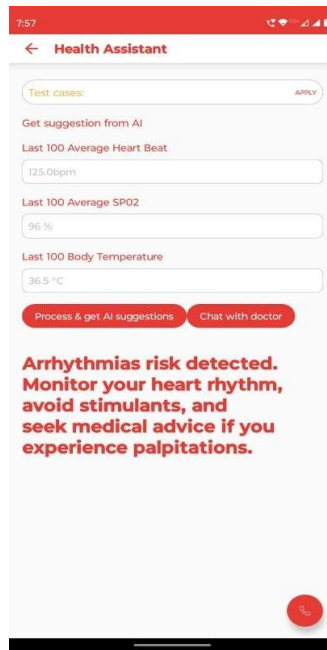


Fig 4.7 Prediction Results of Arrhythmias Risk

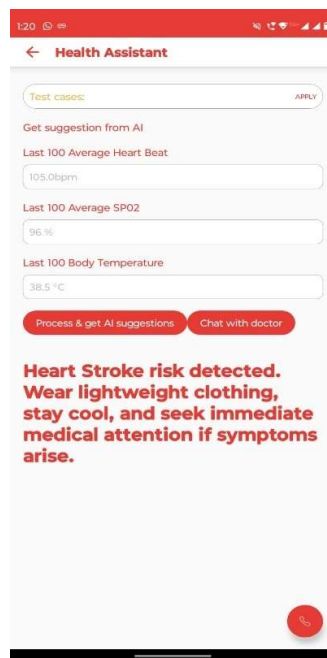


Fig 4.8 Prediction Results of Heart Stroke Risk

CHAPTER V

CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion

This project presents a cost-effective and real-time health monitoring system using IoT technology. The system is built using the Arduino Nano, MAX30100 sensor, LM35 temperature sensor, and HC-05 Bluetooth module. It helps users monitor important health signs like heart rate, blood oxygen level (SpO₂), and body temperature, making it useful for both patients and healthcare providers.

Key Points:

1. **Real-Time Monitoring:** The system collects health data in real time and sends it to a mobile app. This helps users and doctors track health continuously and respond quickly, especially for people with long-term health problems.
2. **Accurate Readings:** The sensors used (MAX30100 and LM35) give accurate measurements, close to those of clinical devices. The Arduino Nano processes this data smoothly and ensures reliable communication with the mobile app.
3. **Wireless Connectivity:** With the help of the HC-05 Bluetooth module, health data is sent wirelessly from the Arduino to the mobile app. This makes the system portable and ideal for use in remote or rural areas.
4. **Easy to Use:** The mobile app is user-friendly and shows data in a clear format. Even people with little or no technical knowledge can use it easily.
5. **Low Power Usage:** The system runs on a 9V battery and can work for over 12 hours, making it suitable for long use without a constant power supply.

5.2 Future Enhancement

Although the current system works well, there are many ways to improve and expand it. Below are the key areas for future development:

Better Sensor Accuracy

- Improve the accuracy of the MAX30100 and LM35 sensors through better calibration methods.
- Use advanced sensors like digital thermometers and high-precision pulse oximeters.
- Add self-calibration to adjust for environmental changes like light and skin contact.

Add More Health Monitoring Features

- Include new sensors to measure glucose levels, ECG, and blood pressure.
- This would allow better monitoring for conditions like diabetes, heart disease, and hypertension.

Cloud Data Storage and Analysis

- Use cloud platforms like Firebase or Google Cloud to store health data.
- Cloud access lets doctors check patient data remotely and track long-term trends.
- Use AI tools to analyze data and detect health problems early.

Improve Communication Range

- Replace the HC-05 with modules like ESP32 (Wi-Fi) or long-range Bluetooth modules (BLE or HC-08).
- Consider using LoRa technology for remote areas with poor network access.

AI and Machine Learning

- Add AI models to detect patterns in health data and predict potential health issues.
- Give users personalized advice and alerts for early intervention.

Mobile App Improvements

- Improve the app by adding graphs, reports, and health tips.
- Allow multiple users to track the same patient's health, useful for caregivers and families.

Data Security and Privacy

- Use encryption and secure data transmission to protect sensitive health information.
- Follow standards like HIPAA or GDPR to ensure data safety and legal compliance.

Telemedicine Integration

- Connect the system to telemedicine platforms so patients can consult doctors remotely.
- This will help those in rural or remote areas get medical support without visiting a hospital.

APPENDICES

APPENDIX 1

SOURCE CODE

```
#include <Wire.h>

#include "MAX30100_PulseOximeter.h"

#define BEAT_LED 12

#define ACT_LED A3

#define REPORTING_PERIOD_MS 1000

PulseOximeter pox;

uint32_t tsLastReport = 0;

short beat_detected = 0;

void onBeatDetected() {

    //Serial.println("Beat!");

    beat_detected = 1;

    digitalWrite(BEAT_LED, HIGH);

}

void setup() {

    Serial.begin(9600);

    pinMode(BEAT_LED, OUTPUT);

    pinMode(ACT_LED, OUTPUT);

    Serial.print("Initializing pulse oximeter..");

    digitalWrite(ACT_LED, LOW);
```

```

digitalWrite(BEAT_LED, LOW);

// Initialize sensor

if (!pox.begin()) {

    Serial.println("FAILED");

    for(;;);

} else {

    Serial.println("SUCCESS");

}

// Configure sensor to use 7.6mA for LED drive

pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);

// Register a callback routine

pox.setOnBeatDetectedCallback(onBeatDetected);

}

float getTemperature(){

    float temp = analogRead(A0);

    temp = temp * 0.488;

    return temp;

}

int sample_counter = 0;

float heart_samples = 0 , hb = 0;

int spo2_samples = 0 , sp = 0;

void loop() {

```

```

// Read from the sensor

pox.update();

// Grab the updated heart rate and SpO2 levels

if (millis() - tsLastReport > REPORTING_PERIOD_MS) {

    if(beat_detected == 1){

        hb = pox.getHeartRate();

        sp = pox.getSpO2();

        /*Serial.print("Heart rate:");

        Serial.print(hb);

        Serial.print("bpm / SpO2:");

        Serial.print(sp);

        Serial.println("%");

        Serial.print("Temperature:");

        Serial.println(getTemperature());*/

        tsLastReport = millis();

        digitalWrite(BEAT_LED, LOW);

        beat_detected = 0;

        heart_samples = heart_samples + hb;

        spo2_samples = spo2_samples + sp;

        sample_counter++;

        if(sample_counter > 20)

            //Serial.println("*****");

```

```

Serial.print("{\hb\");

float h_avg = heart_samples/20;

if(h_avg>50 && h_avg<80)

    h_avg = 85.7;

Serial.print(h_avg);

Serial.print(",\sp\");

float sp_avg = spo2_samples/20;

if(sp_avg > 100)

    sp_avg = 99;

if(sp_avg < 95)

    sp_avg = 98;

Serial.print(sp_avg);

Serial.print(",\temp\");

Serial.print(getTemperature());

Serial.println("{}");

heart_samples = 0;

spo2_samples = 0;

hb = 0;

sp = 0;

sample_counter = 0;

}

}

```

REFERENCES

1. C.-H. Huang and J.-W. Guo, "Design of Reflectance Pulse Oximeter and BPM using the MAX30100 Sensor in Early Detection of Hypoxemia in Patients with Cardiovascular Disorders," *Int. J. Adv. Health Sci. Technol.*, vol. 1, no. 1, pp. 1–8, 2020.
2. A. Vaibhav and I. Ahmad, "IoT-Based Patient Health Monitoring System," in *Microelectronics, Electromagnetics and Telecommunications*, J. Anguera, S. C. Satapathy, V. Bhateja, and K. V. N. Sunitha, Eds. Singapore: Springer, 2018, pp. 167–175.
3. P. Kshirsagar, A. Pote, K. K. Paliwal, V. Hendre, P. Chippalkatti, and N. Dhabekar, "A Review on IoT Based Health Care Monitoring System," in *Proc. Int. Conf. Commun. Comput. Electron. (ICCCE 2019)*, Singapore: Springer, 2020, pp. 113–122.
4. T. Shaik et al., "Remote patient monitoring using artificial intelligence: Current state, applications, and challenges," *arXiv preprint arXiv:2301.10009*, 2023.
5. O. Cheikhrouhou et al., "A Lightweight Blockchain and Fog-enabled Secure Remote Patient Monitoring System," *arXiv preprint arXiv:2301.03551*, 2023.
6. L. Linkous, N. Zohrabi, and S. Abdelwahed, "Health Monitoring in Smart Homes Utilizing Internet of Things," *arXiv preprint arXiv:1910.07058*, 2019.
7. M. Sathya, S. Madhan, and K. Jayanthi, "Internet of Things (IoT) based health monitoring system and challenges," *Int. J. Eng. Technol.*, vol. 7, no. 1.7, pp. 1–5, 2018.
8. V. D. Soni, "An IoT Based Patient Health Monitoring System," *Int. J. Integr. Educ.*, vol. 1, no. 1, pp. 43–48, 2018.

9. K. Perumal and M. Manohar, "A Survey on Internet of Things: Case Studies, Applications, and Future Directions," in *Internet of Things: Novel Advances and Envisioned Applications*, Springer, 2017, pp. 281–297.
10. D. S. R. Krishnan, S. C. Gupta, and T. Choudhury, "An IoT based patient health monitoring system," in *Proc. Int. Conf. Adv. Comput. Commun. Eng. (ICACCE)*, 2018, pp. 1–7.